**Mohammad Abdul Jabbar**

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Accomplished Data Engineer with 3+ years in healthcare, education, and retail domains, skilled in **SQL, Python, and Cloud technologies (AWS and GCP)**. Proven success in enhancing system efficiency and automation, with strong problem-solving and collaboration skills. Experienced in **CI/CD, data warehousing, and real-time analytics**. Excellent communicator dedicated to delivering high-quality solutions.

## Education:

**Master of Science (MS):** Information Technology, Arizona State University, Tempe. **Jan 2023 – Dec 2024**

## Work Experience:

**Data Engineer | OSAZ** **Feb 2025 - Present**

- Developed a machine learning model for skin condition classification using **Python, TensorFlow, and PyTorch**. Applied convolutional neural networks and ensemble techniques to predict pores, roughness, and blackheads from facial images. Improved classification accuracy by **17%** compared to baseline models, reducing manual labeling effort and enhancing diagnostic consistency.
- Built a high-performance **Flask-based API** for skin tone and disease predictions with real-time inference on **Vertex AI**. Integrated caching and asynchronous Cloud Functions to support **95%** of user requests under 300ms, improving system responsiveness and scalability across devices.
- Developed a **React.js** web application delivering personalized shade match and skincare recommendations. Integrated dynamic visualization with user-uploaded images and condition tracking. Achieved a 20% increase in user engagement and 15% improvement in result relevance through iterative A/B testing and feedback loops.
- Spearheaded **CI/CD** automation using **GitHub Actions** and **Docker** for full-stack deployment to **Google Cloud**. Automated test, build, and rollout processes across backend and model services, reducing deployment time by 35% and doubling release frequency, enabling faster ML model iteration and frontend enhancements.

**Data Analyst | Arizona State University** **Aug 2023 – Dec 2024**

- Developed a data-driven dashboard in **Tableau** to monitor and visualize real-time study room occupancy trends across campus libraries. Integrated **SQL-based data pipelines**, enabling staff to make informed decisions and improving space utilization by 20%.
- Automated analysis of booking logs using **Python and SQL**, uncovering a 35% no-show rate during peak hours. This led to the implementation of new reservation policies, improving room availability and fairness among students.
- Built predictive models using **Python (scikit-learn)** to forecast room demand by day and hour, improving scheduling efficiency and reducing booking conflicts by 18%. Improved analytical performance by 15% by refactoring legacy SQL queries with **Common Table Expressions (CTEs) and window functions**, enabling faster aggregation and filtering of 1M+ reservation records.
- Standardized QA and reporting processes by documenting **ETL workflows**, business logic, and metrics in **Confluence**. This enhanced transparency and enabled cross-functional collaboration between the analytics and operations teams.

**Data Analytics Engineer | Infosys | Client: Conagra Brands** **Nov 2020 - Dec 2022**

- Led the transformation of legacy **ETL workflows** into modular data services using **Python and AWS Glue**. Designed and deployed independent data pipelines to improve scalability and resilience, resulting in a 45% reduction in job failure rates and a 30% increase in overall pipeline reliability across Conagra's supply chain systems.
- Improved **API performance by 40%** through implementation of multithreaded data sync processes and intelligent caching for inventory management endpoints. Optimized data retrieval for high-throughput applications, enhancing user experience and reducing report generation time by **50%**.
- Automated infrastructure provisioning on **AWS** using **Python and Terraform**. Built infrastructure-as-code templates for **S3, Lambda, and Glue** orchestration, achieving **99.9%** uptime and reducing manual configuration effort by **25%**, enabling faster onboarding of new data pipelines.
- Mentored junior developers and analysts in **Python, SQL** optimization, and cloud data engineering best practices. Conducted code reviews, pair programming, and knowledge-sharing sessions, improving team productivity and ensuring high standards in pipeline quality and performance delivery.

## Projects:

**Data Processing Enhancement for Spatial Queries (Python, SQL, PostgreSQL)**

- Developed and optimized Scala APIs using Spark and Spark-SQL, leading to a 13% increase in the efficiency of SQL query processing, thereby accelerating data handling in Python and PostgreSQL environments.

**Twitter Data Pipeline (Python, Apache Spark, Apache Airflow, AWS, Git)**

- Developed and deployed a high-volume, real-time Twitter data pipeline on AWS using Apache Airflow and Apache Spark. Extracted, processed, and analyzed Twitter data streams, optimizing data processing efficiency by 30%. Utilized Python, AWS services (e.g., EC2, S3), and Git for development and deployment.

## Technical Skills:

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| **Programming Languages**: | Java, Python, C, C++, Scala, JavaScript, SQL |
| **Big Data**: | Spark, Databricks, Hadoop |
| **Cloud Computing Tools**: | AWS, AWS CLI, AWS Lambda, AWS EC2, GCP Cloud Storage, Cloud Run Functions, Vertex AI |
| **ETL Tools**: | TensorFlow, Scikit-Learn, Airflow, Pyspark, Jenkins, Docker, Kubernetes, GitHub, Agile Methodology |
| **Data Modelling**: | Data Warehousing, Start Schema, Snowflake-Schema Modelling, FACT, Dimensional and Pivot tables |
| **Database**: | PostgreSQL, MySQL, MS SQL, Spark SQL, DynamoDB, AWS RDS, Redshift |
| **Reporting/ Analytical Tools**: | Tableau, Power BI, Google Sheets, MS Excel |
| **Certification:** | AWS Certified – Solutions Architect \| SQL Associate by DataCamp |